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Ben-Zion et al.

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(54) **PROCESS FOR THE PREPARATION OF REMIMAZOLAM AND SOLID STATE FORMS OF REMIMAZOLAM SALTS**

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Related U.S. Application Data

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(30) **Foreign Application Priority Data**

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Sep. 20, 2017	(IN)	201711033273
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(51) **Int. Cl.**

C07D 487/04	(2006.01)
A61P 23/00	(2006.01)
C07D 405/14	(2006.01)

(52) **U.S. Cl.**
CPC ***C07D 487/04*** (2013.01); ***A61P 23/00*** (2018.01); ***C07D 405/14*** (2013.01); *C07B 2200/13* (2013.01)

(58) **Field of Classification Search**
CPC C07D 487/04; C07D 405/14; A61P 23/00
USPC 514/220
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,485,635	B2	2/2009	Feldman et al.
9,156,842	B2	10/2015	Tilbrook et al.
2016/0009680	A1	1/2016	Kawakami et al.

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European communication pursuant to Article 94(3) issued in corresponding Application No. EP 18706147.8 dated Jan. 19, 2021 (4 pages).

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(57) **ABSTRACT**

The present disclosure relates to novel processes for the preparation of short acting benzodiazepines as well as to novel intermediates in this process. More particularly the disclosure relates to processes and intermediates for preparation of Methyl 3-[(4S)-8-bromo-1-methyl-6-(pyridin-2-yl)-4H-imidazo[1,2-a][1,4]benzodiazepin-4-yl]propanoate, commonly known as Remimazolam. The present disclosure also relates to solid state forms of Remimazolam salts, processes for the preparation thereof, pharmaceutical formulations/compositions thereof, and methods of use thereof.

8 Claims, 19 Drawing Sheets

Figure 9: Reminazolam Hydrochloride Form I

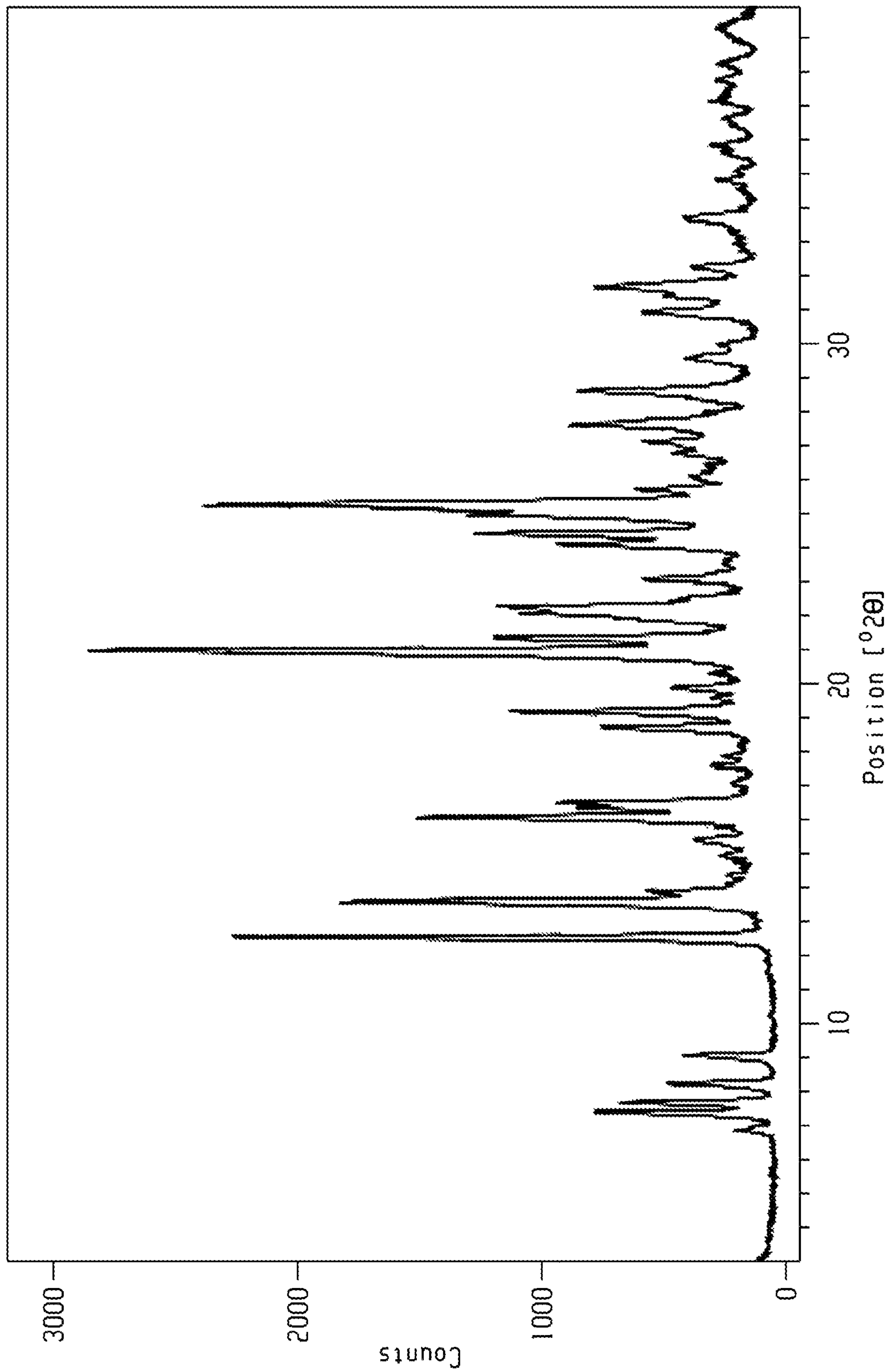
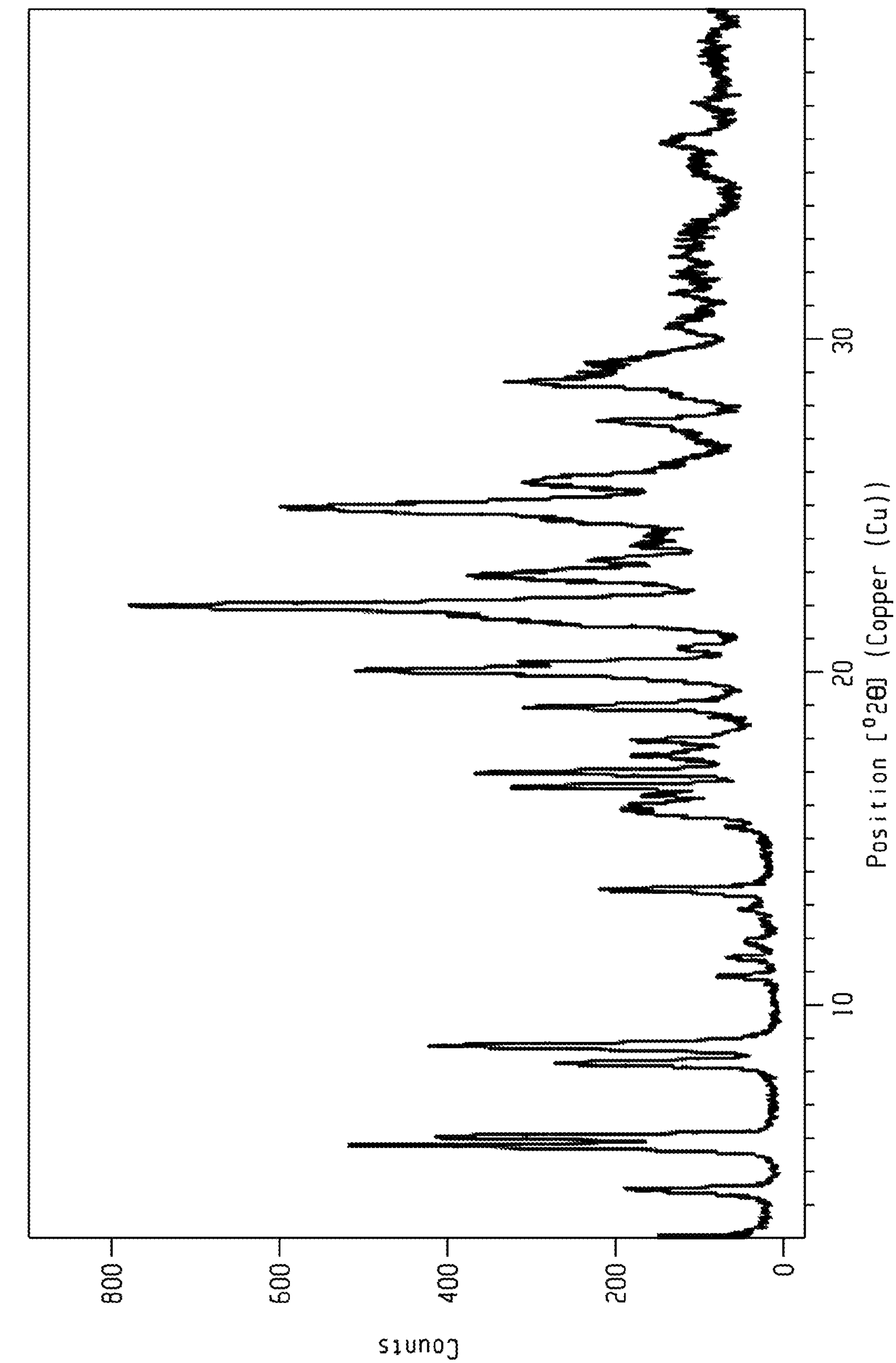


Figure 14: Remimazolam Methane Sulfonate Form 1



PROCESS FOR THE PREPARATION OF REMIMAZOLAM AND SOLID STATE FORMS OF REMIMAZOLAM SALTS

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a divisional of U.S. application Ser. No. 16/484,165, filed Aug. 7, 2019, which is a National Stage of, and claims priority to and the benefit of, International Patent Application No. PCT/US2018/017347 filed Feb. 8, 2018, which, in turn, claims the benefit of, and priority to, Indian Provisional Application No. 201711004706 filed Feb. 9, 2017; Indian Provisional Application No. 201711033273 filed Sep. 20, 2017; Indian Provisional Application No. 201711046934 filed Dec. 27, 2017; and Indian Provisional Application No. 201811002128 filed Jan. 18, 2018, the entire disclosures of each of which are incorporated by reference herein.

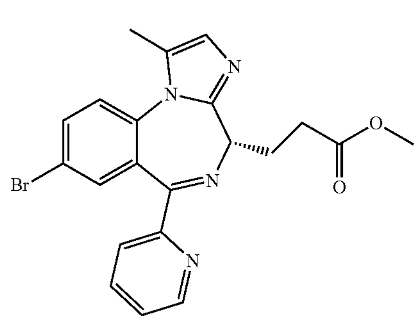
FIELD OF THE INVENTION

The present disclosure relates to novel processes for the preparation of short acting benzodiazepines as well as to novel intermediates in this process. More particularly the disclosure relates to processes and intermediates for preparation of Methyl 3-[(4S)-8-bromo-1-methyl-6-(pyridin-2-yl)-4H-imidazo[1,2-a][1,4]benzodiazepin-4-yl]propanoate, commonly known as Remimazolam. The present disclosure also relates to solid state forms of Remimazolam salts, processes for the preparation thereof, pharmaceutical formulations/compositions thereof, and methods of use thereof.

BACKGROUND

Methyl 3-[(4S)-8-bromo-1-methyl-6-(pyridin-2-yl)-4H-imidazo[1,2-a][1,4]benzodiazepin-4-yl]propanoate, is commonly known as Remimazolam (may referred to herein as Formula-I, Formula I, Compound of formula I or Compound I). Remimazolam has the following chemical structure:

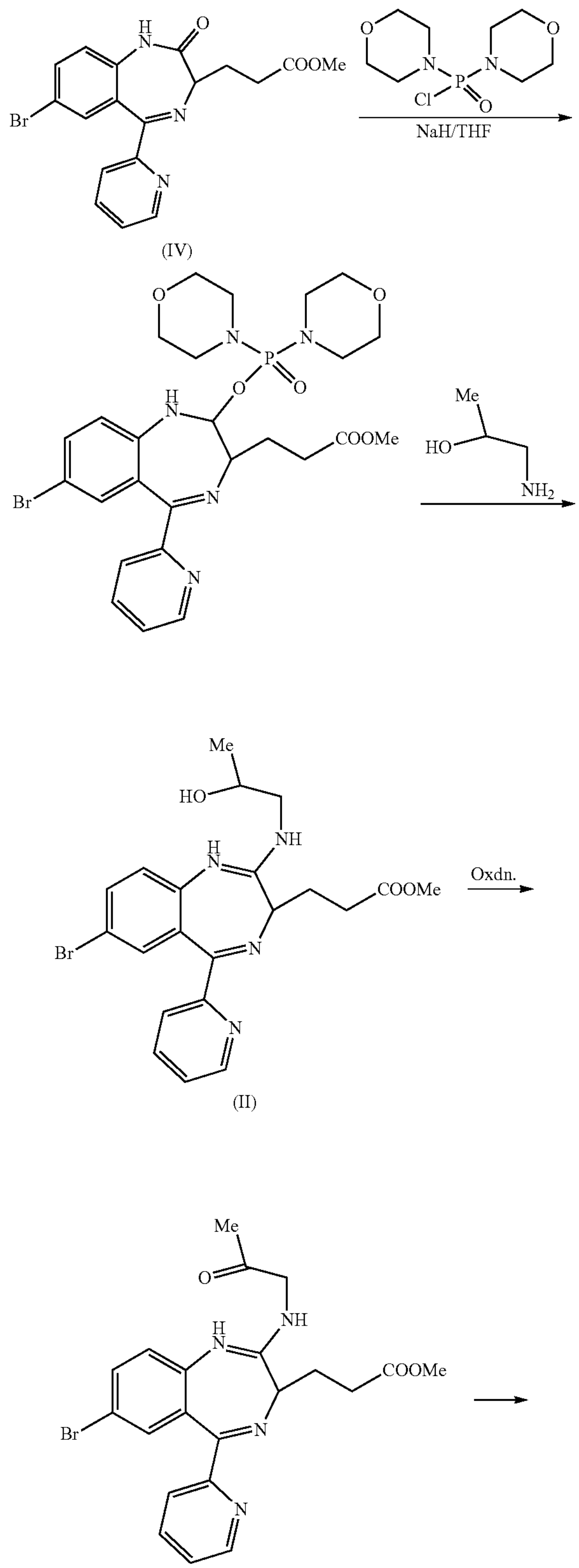
(I)



Remimazolam is developed for use in induction of anesthesia and conscious sedation for minor invasive procedures.

U.S. Pat. No. 7,485,635 (referred to herein as '635) discloses Remimazolam and process for its preparation. The process is described in scheme-1.

Scheme: 1



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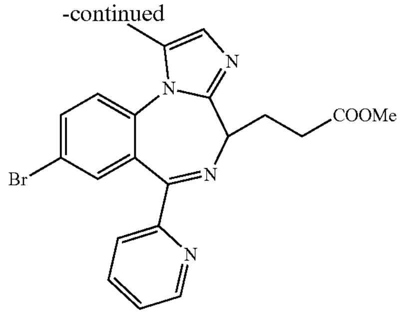


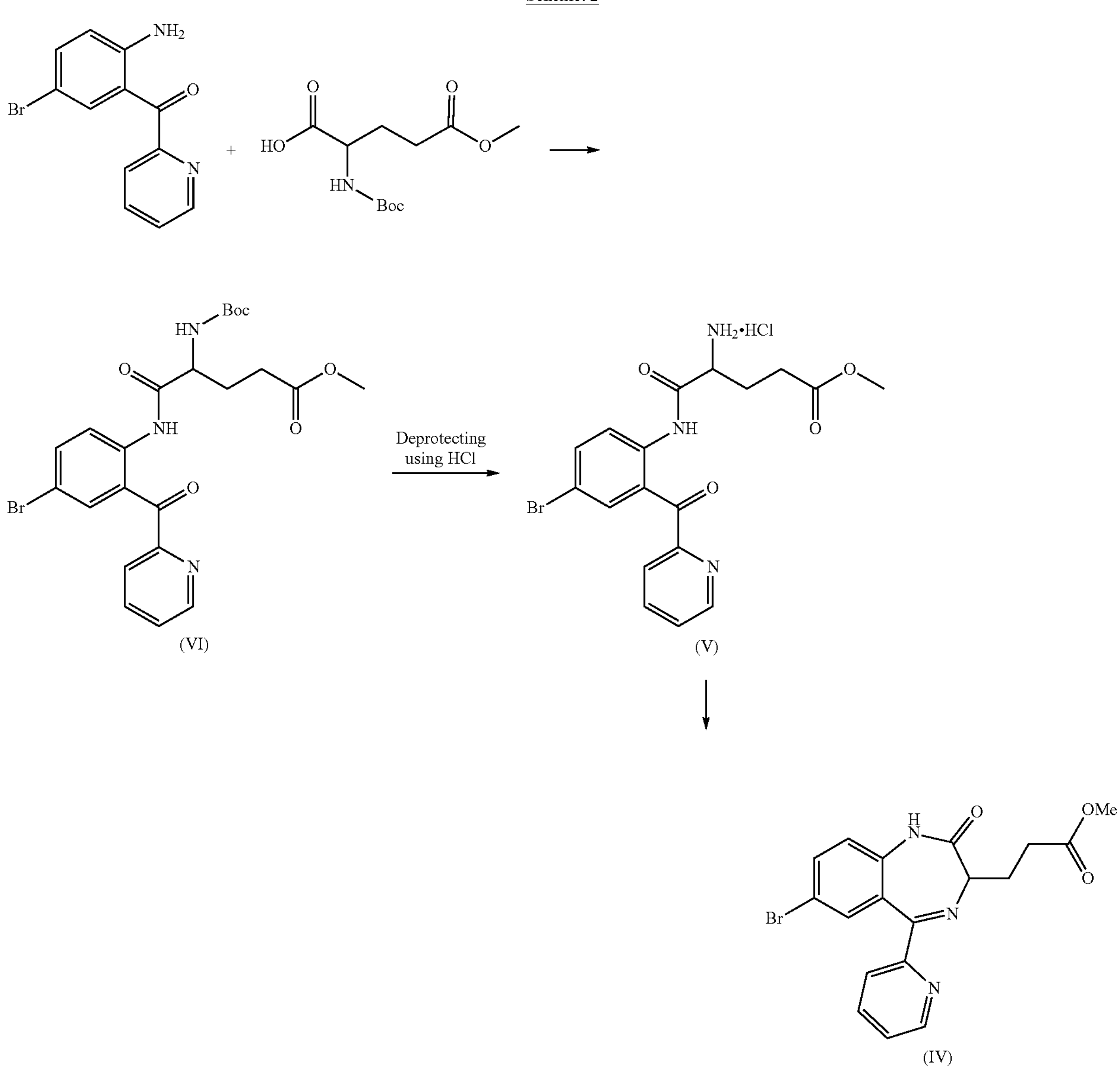
(I)

The process disclosed in '635 patent involves Swern oxidation, which needs to be carried out in the presence of dimethyl sulfoxide (DMSO) and Oxalyl chloride, and requires extremely low temperature of about −78° C., therefore this process can't be adopted for industrial scale manufacturing. In addition the process utilizes the hazardous reagents sodium hydride, Bismorpholinophosphochloridate, and Lawesson's reagent. The reported yield of this process, from intermediate IV to intermediate II is about 37% (w/w) and the overall yield is of about 33%.

U.S. Pat. No. 9,156,842 (referred to herein as '842) discloses a process for preparation of Remimazolam which utilizes Dess-Martin periodinane (DMP) for the oxidation of Compound II, an intermediate in the synthesis of Remimazolam. The reported overall yield is about 34-35% (w/w). The use of DMP is not preferred due to its potential explosive nature, its high cost and the need to use high quantity of reagent to match its high molecular weight.

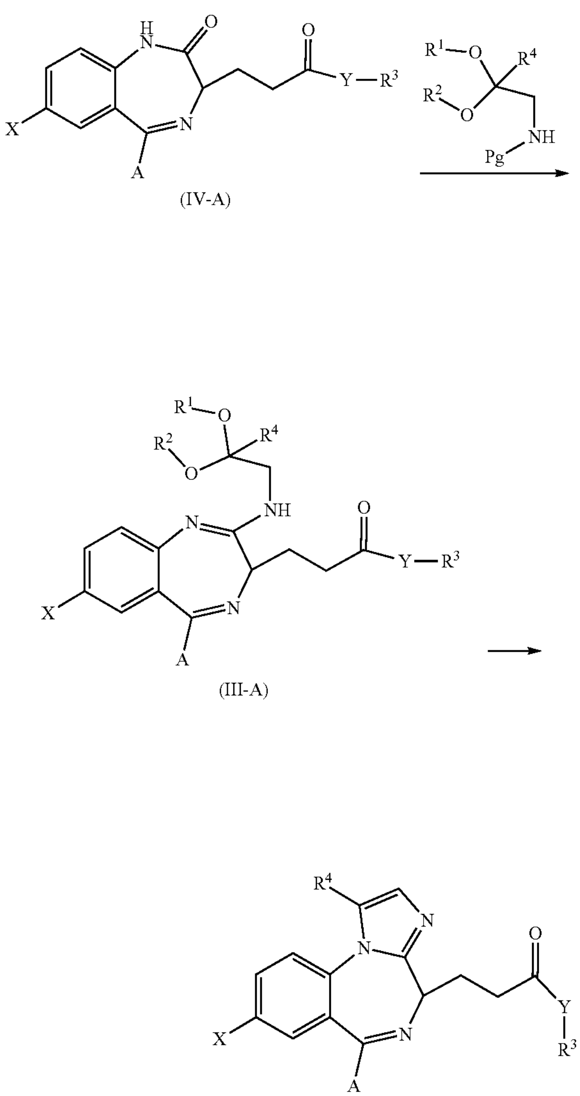
The '842 patent also discloses process for preparation of compound of Compound-IV, an intermediate in the synthesis of Remimazolam, which includes tert-butyloxycarbonyl (Boc) protection and later de-protection using hydrochloric acid. This process is described in Scheme 2.

Scheme: 2



The process can be summarized by the following Scheme-3:

Scheme: 3



wherein, R^1 and R^2 are independently Alkyl or they are connected to form a cyclic ring,

R^3 and R^4 are independently H, alkyl, acyl, vinyl or allyl,

X is any halogen,

Y is O, S or N, or Y is NR^5 wherein R^5 is H, alkyl, acyl, vinyl or allyl,

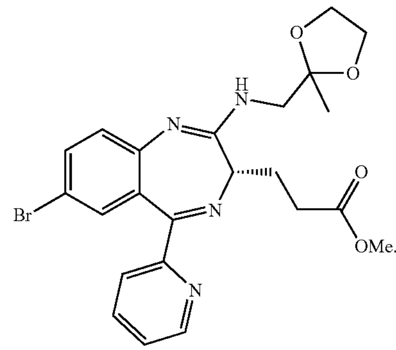
A is alkyl or aliphatic, aromatic or heterocyclic ring; and

Pg is H, amino protecting group or counter ion.

The above process can be done either as a one pot reaction, meaning, the Compound of formula I-A can be prepared from compound of formula IV-A without isolating compound of formula III-A. Alternatively, the compound of formula III-A can be isolated prior to its conversion to compound of formula I-A. Preferably, compound III-A is obtained in a crystalline form.

In specific embodiment, the present disclosure provides novel compound of formula III, its salts, solvates, hydrates or isomers thereof:

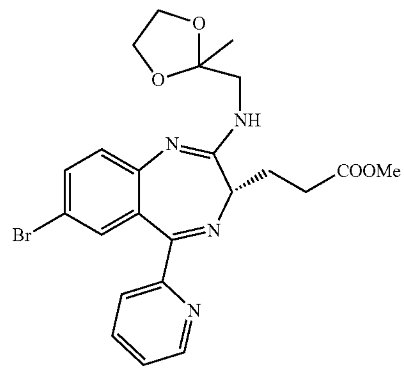
(III)



The compound of formula III can be isolated, either as crystalline or non-crystalline form. Preferably, compound III is crystalline.

The compound of formula III may be a racemic mixture, or enantiomerically pure. Preferably, the compound of formula III is in S-configuration, i.e. compound of formula III-S:

(III-S)



The compound of formula III-S is typically characterized by mass spectra as shown in FIG. **1**, or by ^{1}H-NMR as shown in FIG. **2**, or by combination thereof.

Crystalline compound of formula III-S may be characterized by data selected from one or more of the following: ^{13}C-NMR Spectra of the compound of formula III-S as shown in FIG. **3**; IR Spectra of the compound of formula III-S as shown in FIG. **4**; and combinations of these data.

The compound of Formula-III, its salts, solvates, hydrates or isomer thereof can be prepared by a process comprising preparing the compound of formula-IV and converting it to compound of Formula-III.

The compound of Formula-III, its salts, solvates, or hydrates thereof are useful in the preparation of Remimazolam, the compound of formula I.

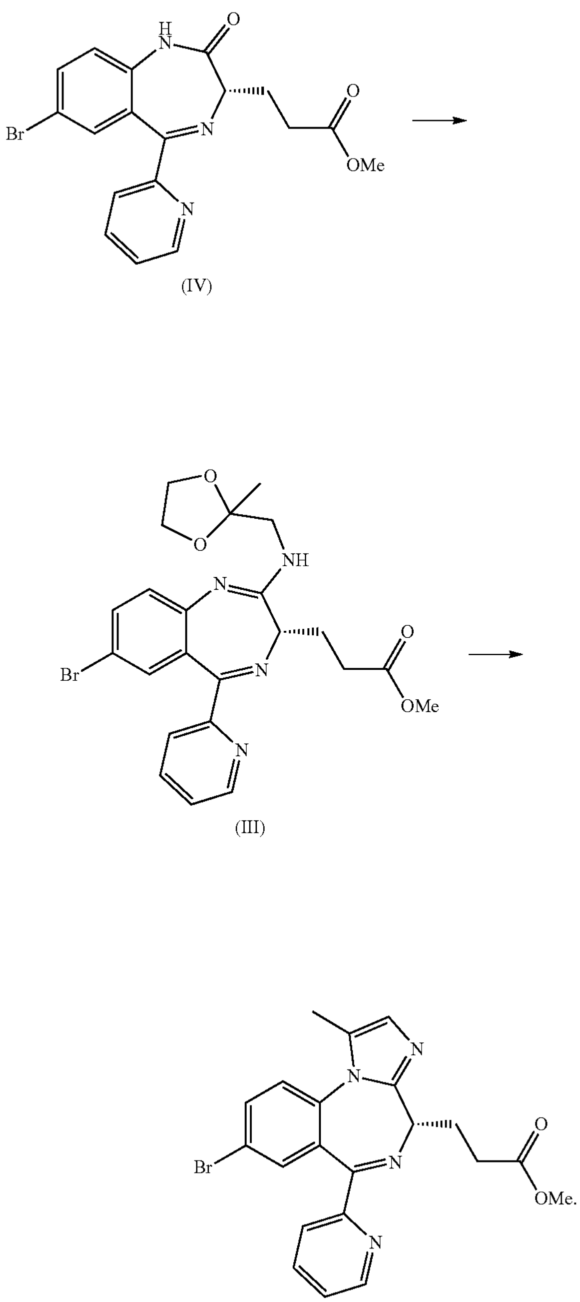
In another embodiment the present disclosure provides a process for preparing Remimazolam comprising:

a. reacting the compound of formula IV and an amino ketal compound in presence of a carbonyl activating group to obtain the compound of formula III; and
b. converting the compound of formula III to the compound of formula I, by deprotecting the ketal followed by cyclization in acidic conditions.

Amino ketal compound and carbonyl activating group are as defined above, preferably the amino ketal is 2-methyl-1,3-dioxolan-2-yl)methanamine. The process can be summarized by the following Scheme-4:

Scheme: 4

Scheme: 4



(IV)

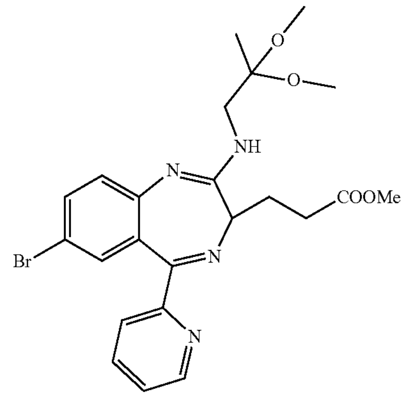
(III)

(I)

The above process can be done either as a one pot reaction, meaning, the compound of formula I can be prepared from compound of formula IV without isolating compound of formula III. Alternatively, the compound of formula III can be isolated prior to its conversion to compound of formula I.

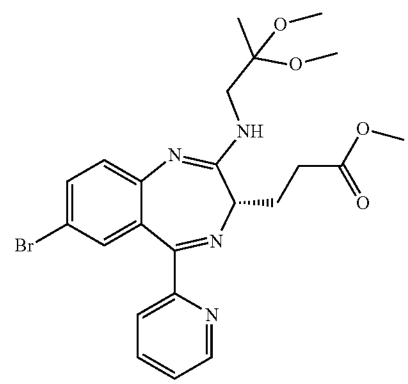
In specific embodiment, the present disclosure provides novel compound of formula III-13, its salts, solvates, hydrates or isomers thereof:

(III-13)



The compound of formula III-13 can be isolated, either as crystalline or non-crystalline form. Preferably, compound III-13 is crystalline.

The compound of formula III-13 may a racemic mixture, or enantiomerically pure. Preferably, the compound of formula III-13 is in S-configuration, i.e. compound of formula III-13 S:



The compound of formula III-13 S is typically characterized by mass spectra as shown in FIG. **5**, or by ^{1}H-NMR as shown in FIG. **6**, or by combination thereof.

Crystalline compound of formula III-13 S may be characterized by data selected from one or more of the following: ^{13}C-NMR Spectra of the compound of formula III-13S as shown in FIG. **7**; IR Spectra of the compound of formula III-13S as shown in FIG. **8**; and combinations of these data.

The compound of Formula-III-13, its salts, solvates, hydrates or isomer thereof can be prepared by a process comprising preparing the compound of formula-IV and converting it to compound of Formula-III-13.

The compound of Formula-III-13, its salts, solvates, or hydrates thereof are useful in the preparation of Remimazolam, the compound of formula I.

In another embodiment the present disclosure provides a process for preparing Remimazolam comprising:

a. reacting the compound of formula IV and an amino ketal compound (such as 1-amino-2,2-dimethoxypropane) in presence of a carbonyl activating group to obtain the compound of formula III-13; and

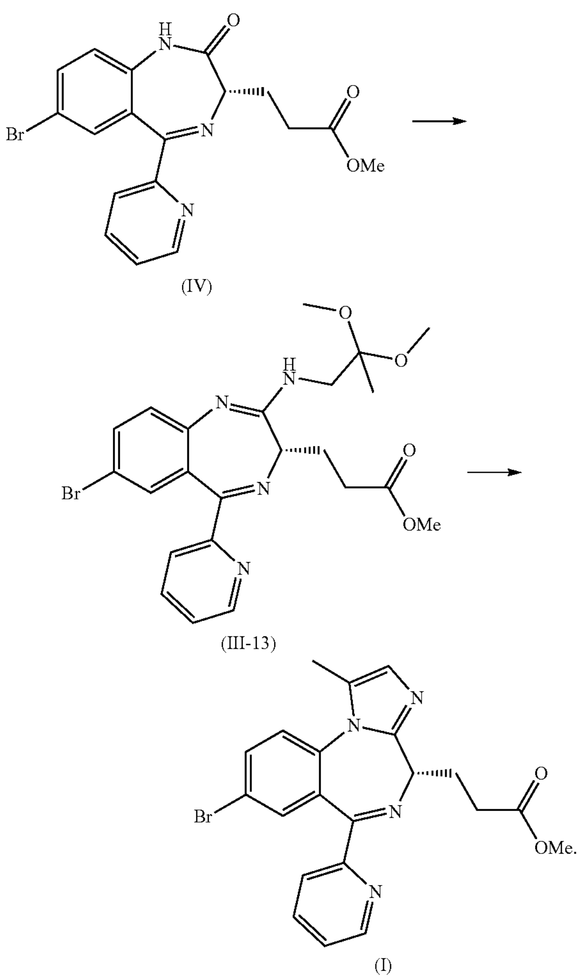
b. converting the compound of formula III-13 to the compound of formula I, by deprotecting the ketal followed by cyclization in acidic conditions.

Amino ketal compound and carbonyl activating group are as defined above.

The process can be summarized by the following Scheme-4a:

Scheme: 4a

Scheme: 4a

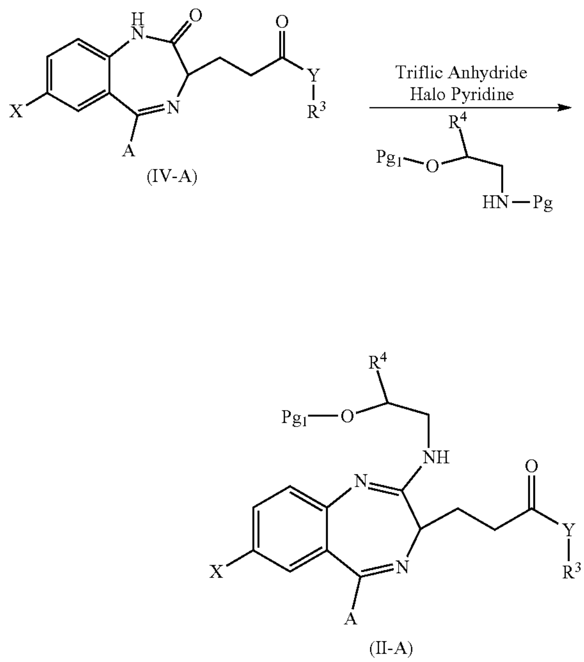


The above process can be done either as a one pot reaction, meaning, the compound of formula I can be prepared from compound of formula IV without isolating compound of formula III-13. Alternatively, the compound of formula III-13 can be isolated prior to its conversion to compound of formula I.

In another embodiment, the present disclosure provides additional processes for preparation of Remimazolam, comprising reacting the compound of formula IV-A and amino-alcohol, such as 1-Aminopropan-2-ol in the presence of triflic anhydride and halopyridine, to form the compound of formula II-A. This process avoids the use of hazardous reagent, can be performed at relatively short reaction time and provides the compound of formula II-A in high yield, of at least 80% (w/w).

The process can be summarized by the following Scheme-5:

Scheme: 5



wherein as R^3 and R^4 are independently H, alkyl, acyl, vinyl or allyl,

X is any halogen,

Y is O, S or N, or Y is NR^5 wherein R^5 is H, alkyl, acyl, vinyl or allyl,

A is alkyl or aliphatic, aromatic or heterocyclic ring, and

Pg_1 is H or any hydroxyl protecting group.

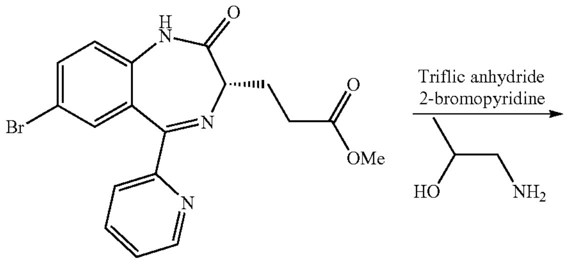
Preferably, as used herein, the term "Protecting group" refers to a grouping of atoms that when attached to a reactive functional group in a molecule masks, reduces or prevents reactivity of the functional group. Examples of protecting groups can be found in Green et al., "Protective Groups in Organic Chemistry", (Wiley, 2nd ed. 1991) and Harrison et al., "Compendium of Synthetic Organic Methods", Vols. 1-8 (John Wiley and Sons, 1971-1996). Representative amino protecting groups include, but are not limited to, formyl, acetyl, trifluoroacetyl, benzyl, benzyloxycarbonyl ("CBZ"), tert-butoxycarbonyl ("BOC"), trimethylsilyl ("TMS"), 2-trimethylsilylethanesulfonyl, ("SES"), trityl and substituted trityl groups, allyloxycarbonyl, 9-fluorenylmethyloxycarbonyl ("FMOC"), nitro-veratryloxycarbonyl ("NVOC") and the like.

Representative hydroxy protecting groups include, but are not limited to, those where the hydroxy group is either acylated or alkylated such as benzyl, and trityl ethers as well as alkyl ethers, tetrahydropyranyl ethers, trialkylsilyl ethers and allyl ethers.

More specifically the present invention relates to the process of preparation of compound of Formula-II from the compound of formula-IV as described in Scheme-6.

Scheme: 6

Scheme: 6



(IV)

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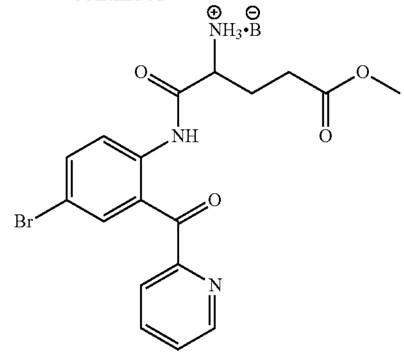
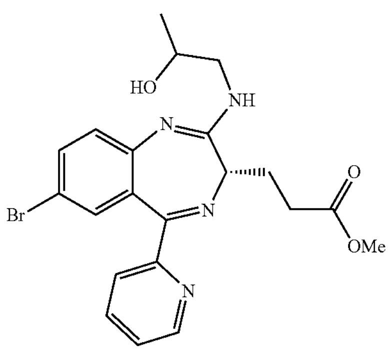


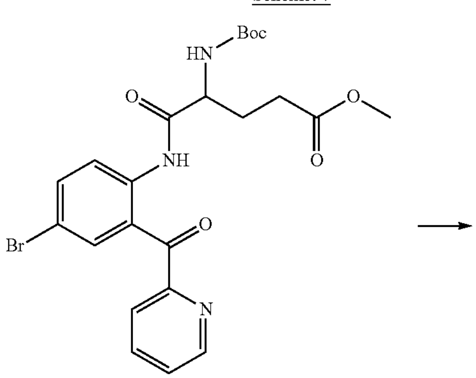
(V-A)



(II)

In yet another embodiment, the present disclosure provides the preparation of early intermediates in the synthesis of Remimazolam, namely the preparation of the compound of formula-V-A. The process comprises de-protection of the compound of formula VI using a suitable acid, excluding hydrochloric acid. The process can be summarized by the following Scheme-7:

Scheme: 7



(VI)

wherein B^- is anion of corresponding acid which is sufficient for the de-protection of -Boc with the proviso that B^- is not Cl^-. Preferably, the acid is triflic acid and B^- is $CF_3SO_3^-$

Remimazolam prepared by the above described processes can be used to prepare solid state forms of Remimazolam, Remimazolam salts and their solid state forms thereof.

The present disclosure comprises a crystalline form of Remimazolam Hydrochloride designated as Form 1. The crystalline Form 1 of Remimazolam Hydrochloride can be characterized by data selected from one or more of the following: an XRPD pattern having peaks at 9.0, 12.5, 13.5, 20.8 and 25.2 degrees two theta±0.2 degrees two theta; an XRPD pattern substantially as depicted in FIG. **9**; or combinations of these data. Crystalline Form 1 of Remimazolam Hydrochloride may be further characterized by data selected from one or more of the following: an XRPD pattern having peaks at 9.0, 12.5, 13.5, 20.8 and 25.2 degrees two theta±0.2 degrees two theta; and also having one, two, three, four or five additional peaks selected from 16.0, 19.1, 19.8, 21.7 and 27.7 degrees two theta±0.2 degrees two theta; or combinations of these data.

The present disclosure further comprises a crystalline form of Remimazolam Dihydrobromide designated as Form 1. The crystalline Form 1 of Remimazolam Dihydrobromide can be characterized by data selected from one or more of the following: an XRPD pattern having peaks at 8.1, 10.3, 12.7, 15.1, 17.0, 20.6, 22.7 and 25.2 degrees two theta±0.2 degrees two theta; an XRPD pattern substantially as depicted in FIG. **10**; or combinations of these data. Crystalline Form 1 of Remimazolam Dihydrobromide may be further characterized by data selected from one or more of the following: an XRPD pattern having peaks at 8.1, 10.3, 12.7, 15.1, 17.0, 20.6, 22.7 and 25.2 degrees two theta±0.2 degrees two theta; and also having one, two, three, four, five or six additional peaks selected from 13.7, 16.6, 17.6, 21.1, 23.5 and 26.6 degrees two theta±0.2 degrees two theta; or combinations of these data.

The present disclosure further comprises a crystalline form of Remimazolam Mono hydrobromide designated as Form 1. The crystalline Form 1 of Remimazolam Mono hydrobromide can be characterized by data selected from one or more of the following: an XRPD pattern having peaks at 7.1, 8.7, 13.6, 22.2 and 24.9 degrees two theta±0.2 degrees two theta; an XRPD pattern substantially as depicted in FIG. **18**; or combinations of these data. Crystalline Form 1 of Remimazolam Mono hydrobromide may be further characterized by data selected from one or more of the following: an XRPD pattern having peaks at 7.1, 8.7, 13.6, 22.2 and 24.9 degrees two theta±0.2 degrees two theta; and also having one, two, three, four or five additional peaks at 20-25° C. The reaction mixture was cooled to 15° C. and slowly charged 10% Methanolic HCl solution (87.0 g) then slowly raised the temperature up to 20-25° C. and maintained under stirring for 3-6 hours to complete the reaction. The reaction mass was cooled to 5-10° C. and quenched with aqueous sodium carbonated solution (13.0 g Sodium carbonate in 200 mL water) and added dichloromethane (100 mL) and stir for 15-20 minutes. The organic layer was extracted in dichloromethane (50 mL×2) and washed with water (50 mL). The organic layer was charged with activated carbon (2.0 g) and maintained for 30 minutes at 35-40° C. Reaction mass was filtered through hyflo bed and washed with hot dichloromethane (20 mL). The clear solution was distilled off under vacuum below 45° C. up to 40-60 ml reaction mass remains inside. Then methyl acetate (60 mL) was added and stripped out 20-30 mL solvent atmospherically below 65° C. The solution obtained was Remimazolam base (Yield: 14.8-16.6 g) with above 95% HPLC purity.

Example 6: Preparation of Compound of Formula-I Besylate Salt

The residue obtained in example-4 (4.4 g) was dissolved in ethyl acetate (40 ml) and a mixture of Benzene sulfonic acid (1.5 g) and methanol (4 ml) at 10° C. to 15° C. was added drop wise to the reaction mixture. The reaction mixture was stirred for about 3 to 4 hours then precipitation occurred. The precipitate was filtered out and washed with Ethyl acetate (10 ml). Finally dried product was obtained (4.7 g).

Example 7: Preparation of Compound of Formula-II from Compound of Formula-IV

The compound of formula-IV (2 g) was dissolved in dichloromethane (20 ml) and the obtained solution was cooled to −5° C. to −10° C. 2-bromo pyridine (1.58 g) was added to the reaction mixture. Triflic anhydride (1.6 g) was slowly added to the reaction mixture at a temperature of below −5° C. The temperature of reaction mixture was maintained below −5° C. for 30 min. Then, 1-Aminopropan-2-ol (0.75 g) and dichloromethane (4 ml) were slowly added mixture while the temperature was maintained below −5° C. for about 30 min. Water was added to the reaction mixture and the layers were separated. The organic layer was distilled off to obtain a residue. The residue was purified with silica column chromatography in Hexane and Ethyl acetate solvents. Product (2.2 g, Yield=96%; HPLC purity=93%) was isolated.

Example 8: Preparation of Compound of Formula-II Hydrochloride Salt from Compound of Formula-IV

The compound of formula-IV (2 g) was dissolved in dichloromethane (20 ml) and the obtained solution was cooled to −5° C. to −10° C. 2-Bromo pyridine (1.58 g) was added to the reaction mixture. Triflic anhydride (1.6 g) was slowly added to the reaction mixture at a temperature below −5° C. The temperature of reaction mixture was maintained below −5° C. for 30 min. Then, 1-Aminopropan-2-ol (0.75 g) and dichloromethane (4 ml) were slowly added to the mixture and temperature was maintained below −5° C. for about 30 min. Water was added to the reaction mixture and the layers separated. The organic layer was distilled off to obtain a residue. The obtained residue was dissolved in ethyl acetate (40 ml) and slowly added methanolic hydrochloride (about 23 ml). The precipitated product was filtered and dried to obtain 2.1 g (85%) as solid product.

Example 9: Preparation of Compound of Formula-V-A from Compound of Formula-VI

The compound of formula-VI (9.4 g) was dissolved in 100 ml toluene at room temperature the obtained solution was cooled to a temperature below 20° C. To this solution, triflic acid (6.4 ml) was added while maintaining the reaction mixture at a temperature of 15 to 10° C. The mixture was stirred for 24 hours at a temperature of 45° C. to 50° C. The solvent was distilled out from the reaction mixture and product was isolated as solid (9 g, yield 90%).

Example 10—Preparation of Remimazolam Hydrochloride Form 1

Remimazolam (4.0 g) was dissolved in ethyl acetate (36.0 mL), and then slowly added into 3.5 mL ethyl acetate-hydrochloric acid (9.0%) solution under stirring in 10 minutes at 20-30° C. which leads to immediate crystallization. The slurry mass was stirred for 60-90 minutes at 20-30° C. Filtered, washed with ethyl acetate (6.0 mL) and was kept under suction in inert atmosphere for 5-10 minutes at 25±5° C. Then, wet cake was dried at 40±5° C. under vacuum for 3 hours. The solid obtained was Remimazolam hydrochloride salt (Yield: 2.5 g).

The XRPD diffractogram of the obtained product is shown in FIG. **9**.

Example 11—Preparation of Remimazolam Dihydrobromide Form 1

Remimazolam (5.0 g) was dissolved in acetone (75.0 mL) and then slowly added 2.0 mL HBr solution (48%) under stirring at 20-25° C. and maintained for 1 hr. The reaction mixture was filtered, washed with acetone (20.0 mL) and suck dried under nitrogen atmosphere. Wet cake was dried under vacuum at 40±5° C. for 3 hours. The solid obtained was Remimazolam dihydromide salt (Yield: 5.3 g).

The XRPD diffractogram of the obtained solid is shown in FIG. **10**.

Example 12—Preparation of Remimazolam Fumarate Form 1

Remimazolam (5.0 g) was dissolved in ethanol (25 mL) at 20-30° C. and then added fumaric acid (1.3 g) under stirring and maintained for 10-15 minutes. The clear solution was cooled down to 0-5° C. in 20-30 minutes and crystallization was observed after 2 hours. The reaction mixture was maintained for 8-10 hours under stirring at 0-5° C., filtered and suck dried under nitrogen atmosphere for 5-10 minutes. Wet cake was dried under vacuum at 40±5° C. for 4 hours. The solid obtained was Remimazolam monofumarate salt (Yield: 1.5 g).

The XRPD diffractogram of the obtained solid is shown in FIG. **11**.

Example 13—Preparation of Remimazolam Oxalate Form 1

Remimazolam (4.0 g) was dissolved in acetone (80 mL) and then added oxalic acid (0.819 g) under stirring at 20-30° C. and maintained for 10 minutes. The reaction mixture was filtered and the clear solution was cooled down to 0-10° C.

and maintained for 60-90 minutes. The crystallization was observed at 0-5° C. after 30 minutes. The solid obtained was filtered, washed with acetone (20 mL), suck dried under nitrogen atmosphere. Wet cake was dried at 40±5° C. under vacuum for 3 h. The solid obtained was Remimazolam oxalate salt. (Yield: 3.1 g).

The XRPD diffractogram of the obtained solid is shown in FIG. **12**.

Example 14—Preparation of Remimazolam Sulphate Form 1

Remimazolam (4.0 g) was dissolved in acetone (80.0 mL) and then slowly added conc. sulphuric acid (0.910 g) under stirring at 20-30° C. which leads to immediate crystallization in 10 minutes. The slurry mass was maintained at 20-30° C. under stirring for 60-90 minutes. Filtered, washed with acetone (20.0 mL) and suck dried under nitrogen atmosphere. Wet cake was dried under vacuum at 40±5° C. for 3 hours. The solid obtained was Remimazolam monosulphate salt (Yield: 4.4 g).

The XRPD diffractogram of the obtained solid is shown in FIG. **13**.

Example 15—Preparation of Remimazolam Methane Sulfonate Form 1

Remimazolam (3.0 g) was dissolved in acetone (60.0 mL) and then slowly added methane sulfonic acid (0.656 g) under stirring at 20-30° C. in 10 minutes which leads to immediate crystallization. The slurry mass was maintained at 20-30° C. under stirring for 60-90 minutes. Filtered, washed with acetone (15.0 mL) and suck dried under nitrogen atmosphere. Wet cake dried at 40±5° C. under vacuum for 4 hours. The solid obtained was Remimazolam methane sulfonate salt (Yield: 3.1 g).

The XRPD diffractogram of the obtained solid is shown in FIG. **14**.

Example 16—Preparation of Remimazolam Camphor Sulfonate Form 1

Remimazolam (3.0 g) was dissolved in acetone (45 ml) and then added R-(−)-Camphor sulfonic acid (1.6 g) under stirring at 20-30° C. which leads to immediate crystallization. The slurry mass was maintained at 20-30° C. under stirring or 60-90 minutes. Filtered, washed with acetone (20 ml) and suck dried under nitrogen atmosphere. Wet cake was dried under vacuum at 40±5° C. for 4 h. The solid obtained was Remimazolam mono-camphor sulfonate salt (Yield: 3.5 g).

The XRPD diffractogram of the obtained solid is shown in FIG. **15**.

Example 17—Preparation of Remimazolam Dibesylate Form 1

Remimazolam besylate Form 1 (1.0 g) was taken in a 50 mL round bottom flask and dissolved in ethanol (14 mL) at 40-45° C. The reaction mixture was filtered to remove any undissolved particulate and a clear solution was obtained. Subsequently, ethanolic solution of benzenesulfonic acid (0.34 g of benzenesulfonic acid in 7.0 mL ethanol) was added under stirring and maintained at 40-45° C. for 1 hour. The clear solution was slowly cooled down to 20-25° C. for 4 hours. The reaction mixture was further cooled to 0-10° C. in 1 hour and maintained under stirring for 16-18 hours, then filtered and suck dried for 5-10 minutes. The solid obtained was Remimazolam dibesylate salt (Yield: 0.65 g). Remimazolam dibesylate salt Form 1 is hydrate.

The XRPD diffractogram of the obtained solid is shown in FIG. **16**.

Example 18—Preparation of Remimazolam Camphor Sulfonate Form 2

Remimazolam (20.0 g) was dissolved in acetone (300 mL) in 500 ml round bottom flask then added R-(−)-Camphor sulfonic acid (10.5 g) at 20° C.-25° C. The obtained slurry mass were maintained under stirring for 2 hours at 20-25° hrs at 200-300 rpm. The reaction mixture was filtered, washed twice with acetone (60 mL) and suck dried the material under vacuum by blanketing with nitrogen atmosphere for 15 min at 20-25° C. The wet cake was dried under vacuum at 45° C. for 8 hr. The solid obtained was Remimazolam mono-camphor sulfonate salt (Yield: 18 g).

The XRPD diffractogram of the obtained solid is shown in FIG. **17**.

Example 19—Preparation of Remimazolam HBR Form 1

Remimazolam (20.0 g) was dissolved in acetone (300 mL) in 500 ml round bottom flask and the obtained solution were stirred at 200-300 rpm for 5 to 10 minutes at 20-25° C. and then aqueous solution of HBr (48%) was added in 20 min under stirring at 20-25° C. and maintained for 2 hours at 20-25° C. The reaction mixture was filtered, washed twice with acetone (40 mL) and suck dried under vacuum by blanketing with nitrogen atmosphere for 15 minutes at 20-25° C. The wet cake was dried under vacuum at 45° C. for 8 hours. The solid obtained was Remimazolam mono hydrobromide salt (Yield: 17.8 g).

The XRPD diffractogram of the obtained solid is shown in FIG. **18**.

Example 20—Preparation of Remimazolam Methane Sulfonate Form 2

Remimazolam (20.0 g) was dissolved in acetone (400 mL) in 500 ml round bottom flask then added methane sulfonic acid (4.3 g) and the obtained slurry mass were maintained under stirring at 20-25° C. for 2 hours at 200-300 rpm. The reaction mixture was filtered, washed twice with acetone (50 mL) and suck dried under vacuum by blanketing with nitrogen atmosphere for 15 minutes at 20-25° C. Wet cake was dried under vacuum at 45° C. for 8 hr. The solid obtained was Remimazolam methane sulfonic acid salt (Yield: 18.5 g).

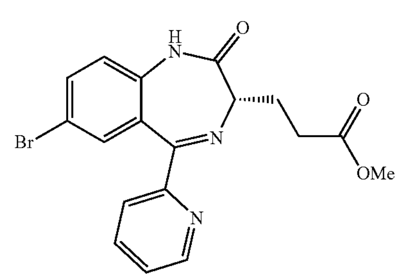
The XRPD diffractogram of the obtained solid is shown in FIG. **19**.

The invention claimed is:

1. A process for preparation of Remimazolam comprising:

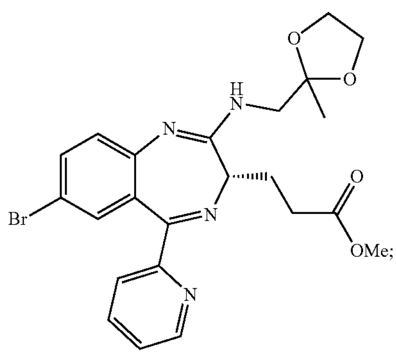
a. reacting a compound of formula IV:

(IV)



and an amino ketal compound selected from (2-methyl-1,3-dioxolan-2-yl)methanamine, (2,5,5-trimethyl-1,3-dioxan-2-yl)methanamine or 1-amino-2,2-dimethoxypropane, in presence of a carbonyl activating group selected from triflic anhydride, triflic acid, benzene sulfonic acid, Toluene sulfonic acid, or Methane sulfonic acid, to obtain a compound of formula III:

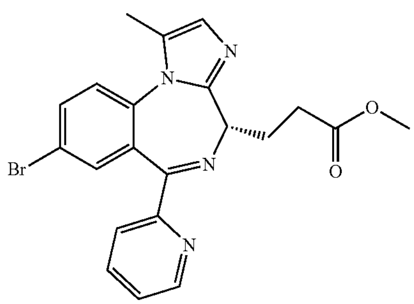
(III)



and

b. converting the compound of formula III to Remimazolam having the following formula

(I)



by deprotecting the ketal followed by cyclization in acidic conditions.

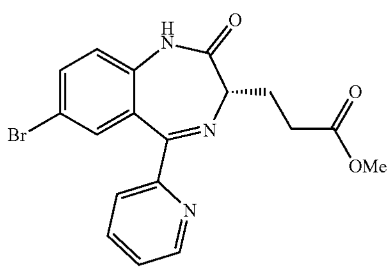
2. The process according to claim 1, wherein the amino ketal is 2-methyl-1,3-dioxolan-2-yl)methanamine.

3. The process according to claim 1, wherein the process is a one-pot process.

4. A process for preparation of Remimazolam comprising:

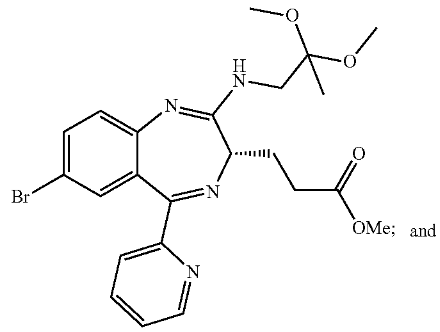
a. reacting a compound of formula IV:

(IV)



and an amino ketal compound selected from (2-methyl-1,3-dioxolan-2-yl)methanamine, (2,5,5-trimethyl-1,3-dioxan-2-yl)methanamine or 1-amino-2,2-dimethoxypropane, in presence of a carbonyl activating group selected from triflic anhydride, triflic acid, benzene sulfonic acid, Toluene sulfonic acid, or Methane sulfonic acid, to obtain a compound of formula III-13:

(III-13)



b. converting the compound of formula III-13 to Remimazolam by deprotecting the ketal followed by cyclization in acidic conditions.

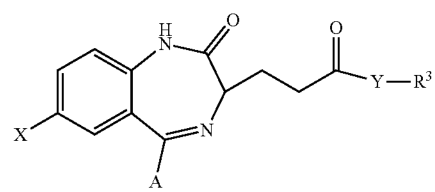
5. The process according to claim 4, wherein the amino ketal is 1-amino-2,2-dimethoxypropane.

6. The process according to claim 4, wherein the process is one-pot process.

7. A process comprising:

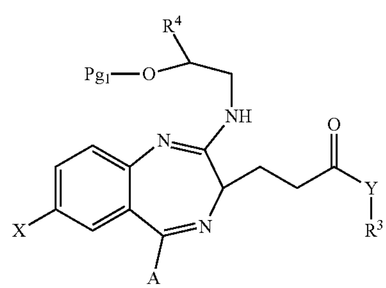
a. reacting a compound of formula IV-A:

(IV-A)



and amino-alcohol in the presence of triflic anhydride and halopyridine to obtain a compound of formula II-A of the following structure:

(II-A)



wherein R^3 is C_1 alkyl, and R^4 is H, alkyl, acyl, vinyl or allyl,

X is Br,

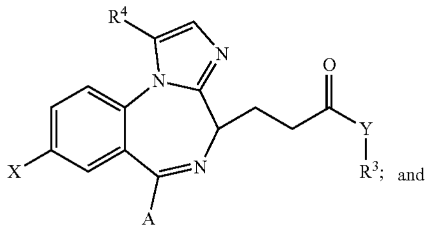
Y is O,

A is 2 pyridinyl, and

Pg_1 is H or any hydroxyl protecting group;

b. converting the compound of formula II-A to a compound of formula I-A:

(I-A)



c. optionally, converting the compound of formula I-A to Remimazolam.

8. The process according to claim **7**, wherein the amino alcohol is 1-Aminopropan-2-ol.

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